

Doping Stability Report -- Zn in KCoO2 at 200 C

Generated: 2026-05-30 16:35 Host: KCoO2 Dopant: Zn

Executive Summary

Zn preferentially occupies the Co layer. The formation energy difference between the two layers is +3.083 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00162 Å²/ps (stable), compared to -0.00663 Å²/ps on the K site. The K-site E_f is approximate due to a charge surplus of +1 that cannot be modelled in CHGNet.

Site	E _f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	-0.216	-1	No	PASS	FAIL	STRUCTURAL COLLAPSE
K	+2.867	+1	Yes	PASS	FAIL	STRUCTURAL COLLAPSE

Zn at the Co layer -- STRUCTURAL COLLAPSE

Charge situation

Zn replaces Co with a charge mismatch of -1. To restore charge neutrality, 1 Na atom(s) were removed from the supercell (furthest from the dopant site). The formation energy accounts for the Na-vacancy term.

Formation Energy (thermodynamic stability)

Formation energy E _f	-0.2162 eV	PASS	< 1.0 eV to pass
---------------------------------	------------	------	------------------

E_f = -0.216 eV. This value is slightly negative, indicating that Zn incorporation is thermodynamically favorable at the Co site.

Molecular Dynamics Stability (200 C, 25 ps)

MSD slope (late-time)	0.00162 Å ² /ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.050 Å ²	----	
Max displacement	0.749 Å	----	
Coordination (min/mean/max)	3 / 4.5 / 5	FAIL	relaxed = 5, tolerance +/-1
Mean bond length (MD vs relaxed)	2.080 Å (relaxed: 2.1)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = 0.00162 Å²/ps. The slope is small and below the migration threshold -- the dopant is site-stable. Final MSD = 0.050 Å²; maximum displacement from starting position = 0.75 Å.

Coordination fluctuated between 3 and 5 (mean 4.5), deviating by more than +/-1 from the relaxed value (5). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.080 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STRUCTURAL COLLAPSE

Doping Stability Report -- Zn in KCoO₂ at 200 C

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Zn atom barely moves (MSD = 0.050 Å²) but the coordination count fluctuates because the bond length at the Co site is close to the analysis cutoff. Consider increasing coordination_cutoff_A and re-running analysis.

? **WARNING:** Charge deficit (-1) cannot be compensated in CHGNet (would require adding K interstitials or oxidising Co). Running single substitution without compensation. E_f is approximate.

Zn at the K layer -- STRUCTURAL COLLAPSE

Charge situation

Zn replaces K with a charge surplus of +1. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E _f	+2.8667 eV	FAIL	< 1.0 eV to pass
---------------------------------	-------------------	-------------	------------------

E_f = +2.867 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Zn incorporation is thermodynamically unfavorable under equilibrium conditions at the K site.

Molecular Dynamics Stability (200 C, 25 ps)

MSD slope (late-time)	-0.00663 Å²/ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.587 Å²	----	
Max displacement	1.068 Å	----	
Coordination (min/mean/max)	2 / 3.9 / 5	FAIL	relaxed = 5, tolerance +/-1
Mean bond length (MD vs relaxed)	2.068 Å (relaxed: 2.1)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = -0.00663 Å²/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.587 Å²; maximum displacement from starting position = 1.07 Å.

Coordination fluctuated between 2 and 5 (mean 3.9), deviating by more than +/-1 from the relaxed value (5). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.068 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STRUCTURAL COLLAPSE

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Zn atom barely moves (MSD = 0.587 Å²) but the coordination count fluctuates because the bond length at the K site is close to the analysis cutoff. Consider increasing coordination_cutoff_A and re-running analysis.

Site Preference Comparison

The Co site is preferred by 3.083 eV over the K site. Formation energy: -0.216 eV (Co) vs +2.867 eV (K).

Doping Stability Report -- Zn in KCoO₂ at 200 C

Zn preferentially occupies the Co layer. The formation energy difference between the two layers is +3.083 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00162 Å²/ps (stable), compared to -0.00663 Å²/ps on the K site. The K-site E_f is approximate due to a charge surplus of +1 that cannot be modelled in CHGNet.

Caveats and Limitations

? CHGNet is charge-neutral.

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch $\neq 0$) are approximate even when structural compensation defects are included.

? No Freysoldt/Kumagai correction.

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? Metal-rich reference state.

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? MD timescales are short (25 ps default).

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? Energy rankings are more trustworthy than absolute values.

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.